

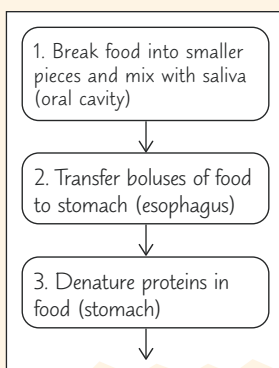
41 Animal Nutrition

KEY CONCEPTS

- 41.1** An animal's diet must supply chemical energy, organic building blocks, and essential nutrients *p. 899*
- 41.2** Food processing involves ingestion, digestion, absorption, and elimination *p. 902*
- 41.3** Organs specialized for sequential stages of food processing form the mammalian digestive system *p. 905*
- 41.4** Evolutionary adaptations of vertebrate digestive systems correlate with diet *p. 911*
- 41.5** Feedback circuits regulate digestion, energy storage, and appetite *p. 915*

Study Tip

Make a flowchart: Digestion occurs stepwise in the human alimentary canal. As you study the molecular



details, keep track of the overall process by adding to this start of a flowchart listing the overall effect and location of each step.

➔ Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 41
- Figure 41.10 Walkthrough: The Stomach and Its Secretions
- BioFlix® Animation: Homeostasis: Regulating Blood Sugar

For Instructors to Assign (in Item Library)

- Tutorial: Fat Absorption and Fat Structure
- Tutorial: What Role Do Genes Play in Appetite Regulation?

Ready-to-Go Teaching Module
(in Instructor Resources)

- The Human Digestive System



Figure 41.1 For this herring gull, dinnertime has arrived. Once this meal has been taken in, the tissues of the sea star will be taken apart and its nutrients taken up. Paradoxically, the classes of nutrients in the sea star—largely proteins, fats, and carbohydrates—also make up the tissues of the gull.

How can animals extract the nutrients they need from food while not digesting their own tissues?

An animal digests food using **compartmentalized processing** in a tube-like system. Compartmentalization protects body tissues while allowing enzymes and acids to break down nutrients.

